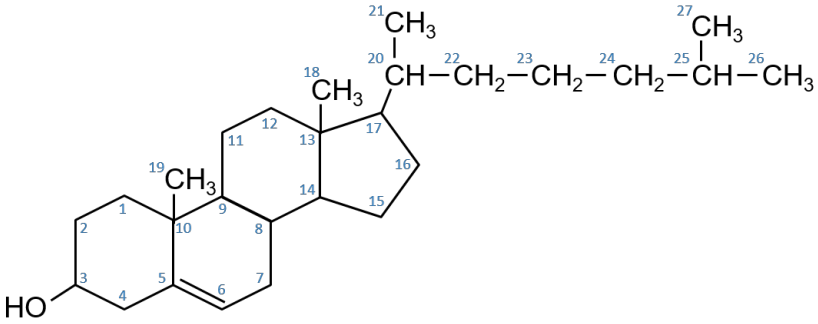
Cholesterol metabolism

Cholesterol is one of the most popular molecules in human biology for its association with **cardiovascular diseases**.

Cholesterol is a lipid that serves as a **stabilizing component of cell membranes** and as a **precursor of many other molecules**, such as bile acids, steroid hormones or cholecalciferol (vitamin D_3). It is obtained from the diet or synthesized in our cells (mostly in the liver and the intestine).

Cholesterol structure

Cholesterol is a **steroid lipid** composed of four fused rings (a structure known as cyclopentaneperhydrophenanthrene or steran) with several substitutions. It contains a **small polar head** formed by a hydroxyl group and a **large nonpolar region** formed by the fused rings and hydrocarbon side chain. Thus, cholesterol is mostly a **hydrophobic molecule with very low solubility in water**. For this reason, it is transported in the blood in particular structures known as **lipoproteins**.



Intestinal absorption and excretion of cholesterol

Dietary cholesterol is taken up from the intestinal lumen into enterocytes mainly through **NPC1L1**, while the ABCG5/ABCG8 transporter, a member of the ATP-binding cassette (ABC) protein family, promotes the efflux of cholesterol and other sterols back into the intestinal lumen, thereby limiting their net absorption. There are drugs that can interfere with cholesterol absorption. Ezetimibe reduces cholesterol absorption by inhibiting NPC1L1.

Elimination of endogenous cholesterol from our body is not an easy task because the ring structure cannot be metabolized. Fecal excretion of secreted bile acids and biliary cholesterol is the principal route of cholesterol disposal.

Cholesterol synthesis

Our body can synthesize cholesterol. The rate of cholesterol synthesis is highly dependent on the amount of dietary cholesterol entering per day. An adult on a low-cholesterol diet can synthesize close to 1 g of cholesterol per day. The **synthesis of cholesterol takes place primarily in the liver**, although the intestine can also synthesize significant amounts. The adrenal cortex and the gonads also synthesize cholesterol, where it is used as a precursor for the synthesis of steroid hormones.

The precursor for cholesterol synthesis is acetyl-CoA. All 27 carbon atoms of cholesterol are ultimately derived from the acetyl groups of acetyl-CoA. Acetyl-CoA can be obtained from several sources, such as β-oxidation of fatty acids, degradation of ketogenic amino acids, or oxidation of pyruvate. Cholesterol synthesis takes place in the cytosol and the smooth endoplasmic reticulum.

The synthesis of cholesterol takes place in 4 stages (the number of carbon atoms per molecule is indicated in parenthesis):

1. Synthesis of mevalonate by the condensation of 3 acetyl-CoA molecules:

   3 Acetyl-CoA (2C each) → mevalonate (6C)

2. Conversion of mevalonate into an activated isoprene unit (a CO_2 molecule is released):

   Mevalonate (6C) → activated isoprene unit (5C)

3. Condensation of 6 activated isoprene units to form squalene:

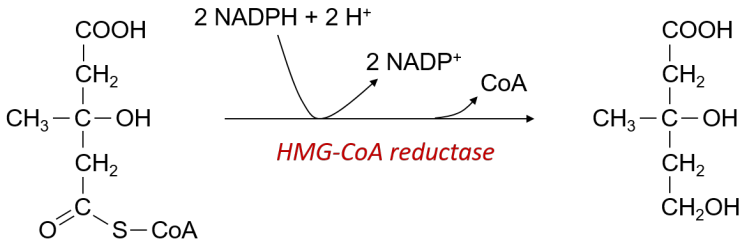
   6 Activated isoprene units (5C each) → squalene (30C)

4. Conversion of squalene to the four-ring steroid nucleus, and subsequent modifications:

   Squalene (30C) → cholesterol (27C)

1. Synthesis of mevalonate

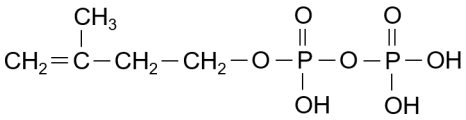
The synthesis of **mevalonate** takes place in the cytosol. Because mitochondrial acetyl-CoA cannot cross the inner mitochondrial membrane, its carbon units are exported mainly as citrate and regenerated as cytosolic acetyl-CoA. The synthesis of mevalonate requires three steps. In the first two steps, three molecules of acetyl-CoA condense to form β-hydroxy-β-methylglutaryl coenzyme A (HMG-CoA). In the third step, mevalonate is produced. **The formation of mevalonate from HMG-CoA is the main rate-limiting step and the checkpoint of cholesterol synthesis**. This reaction is irreversibly catalyzed by ***HMG-CoA reductase***, an integral membrane protein of the smooth endoplasmic reticulum.



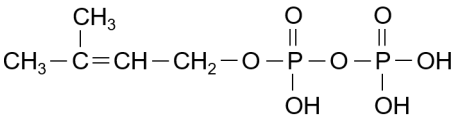
HMG-CoA Mevalonate

2. Conversion of mevalonate into activated isoprene units

Mevalonate is phosphorylated and decarboxylated to produce an activated **isoprene** unit. Three ATP molecules are consumed in the process. The activated isoprene units exist as two interconvertible isomers: isopentenyl diphosphate and dimethylallyl diphosphate.



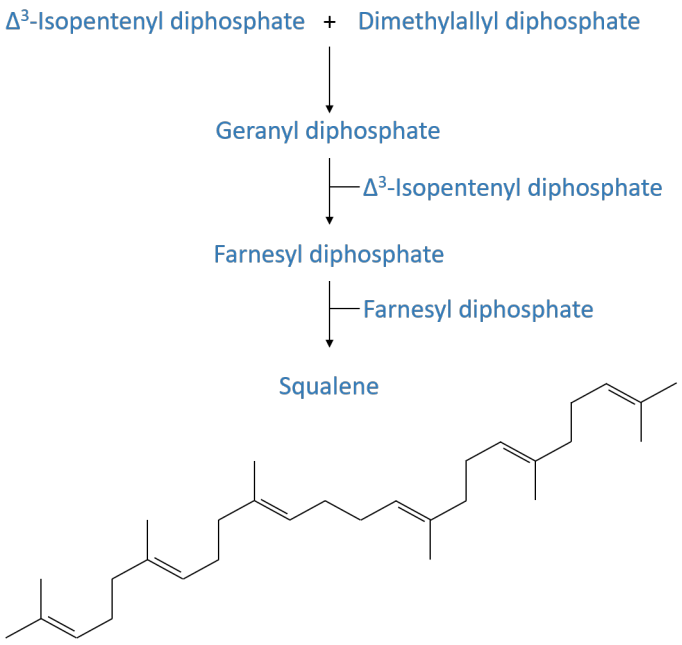
Δ^3-Isopentenyl diphosphate



Dimethylallyl diphosphate

3. Condensation of activated isoprene units

Six activated isoprene units condense to form a molecule of 30 carbon atoms known as **squalene**. The last reaction in the formation of squalene is catalyzed by *squalene synthase*, an enzyme found in the endoplasmic reticulum.



4. Conversion of squalene to cholesterol

By the action of several enzymes, including *monooxygenases*, *cyclases*, *isomerases*, *reductases*, etc., cholesterol is generated from squalene in the endoplasmic reticulum. Cholesterol is the major sterol in vertebrates. Plants predominantly synthesize phytosterols such as sitosterol and stigmasterol, whereas fungi mainly synthesize ergosterol. Bacteria rarely synthesize steroid-like structures.

Requirements for cholesterol synthesis

Cholesterol synthesis is a costly process for cells in terms of energy. Starting from cytosolic acetyl-CoA, the pathway directly consumes 18 **ATP** during the activation of six mevalonate molecules. Higher estimates may include the energetic cost of supplying cytosolic acetyl-CoA. The pathway also requires substantial reducing power in the form of **NADPH**.

Regulation of cholesterol synthesis

The amount of endogenous cholesterol that is synthesized in the liver is tightly regulated by dietary cholesterol levels. The reaction catalyzed by ***HMG-CoA reductase*** **is the main regulatory point for cholesterol synthesis**. The activity of this enzyme, which is highly sensitive to intracellular cholesterol levels, is controlled in multiple ways, including long-term regulation of transcription and degradation, and short-term covalent modification.

Transcriptional regulation of *HMG-CoA reductase*

The rate of **synthesis** of the *HMG-CoA reductase* mRNA is controlled by sterol regulatory element-binding proteins (**SREBPs**). SREBPs are integral proteins of the endoplasmic reticulum.

When ER membrane cholesterol levels decrease, the SREBP-SCAP complex moves from the ER to the Golgi, where SREBP is proteolytically cleaved, releasing its DNA-binding domain. From there, this domain travels to the nucleus and acts as a transcription factor that binds to sterol-regulatory elements (**SRE**) of many genes associated with cholesterol metabolism, including the *HMG-CoA reductase* gene.

Regulation of *HMG-CoA reductase* degradation

HMG-CoA reductase has sterol-sensing regions. When sterol levels increase, conformational changes are induced, which trigger the attachment of ubiquitin molecules and render the enzyme more susceptible to **proteasome degradation**.

Covalent regulation of *HMG-CoA reductase*

The activity of ***HMG-CoA reductase* is inhibited by phosphorylation**. AMP-activated protein kinase (**AMPK**), which senses high AMP levels (i.e. low levels of ATP), catalyzes the phosphorylation of the reductase, thereby reducing its activity. Therefore, when ATP levels are low, cholesterol synthesis decreases.

The enzyme is also regulated by **insulin** and **glucagon**. Insulin induces the dephosphorylation and activation of *HMG-CoA reductase*, whereas glucagon promotes its phosphorylation and inactivation. As a result, **insulin induces cholesterol synthesis, and glucagon inhibits it**.

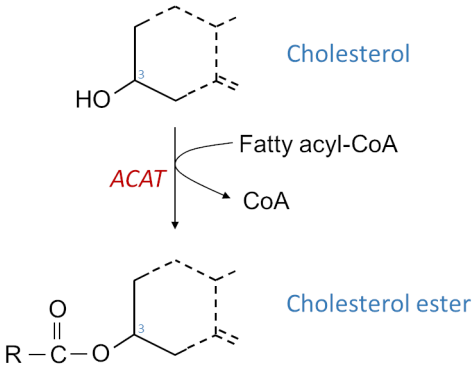
Pharmacological regulation of *HMG-CoA reductase*

Statins, which include natural fungal products and synthetic derivatives, are competitive inhibitors of *HMG-CoA reductase*. They reduce cholesterol synthesis and the production of other mevalonate-derived compounds and are widely used to lower plasma cholesterol levels.

Fates of cholesterol

Only a fraction of hepatic cholesterol is used for the synthesis of hepatic membranes. The majority of cholesterol is exported from the liver as cholesteryl esters, biliary cholesterol or bile acids.

Cholesteryl esters are formed in a reaction catalyzed by *acyl-CoA:cholesterol acyltransferase*. In this reaction, a fatty acid is transferred from coenzyme A to cholesterol (to the hydroxyl group on carbon 3).

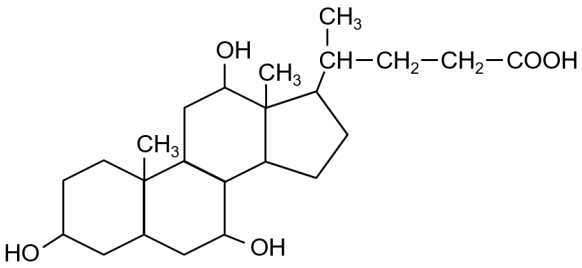


Hepatic cholesteryl esters can be stored in lipid droplets or incorporated into VLDL particles for secretion.

Bile acids

Conversion into **bile acids** is a major route for cholesterol catabolism and elimination. Bile acids, or their derivatives **bile salts**, act as **detergents for emulsifying dietary lipids** in the intestine.

The most abundant bile acids produced in humans are cholic acid and chenodeoxycholic acid. They are easily recognized by the carboxylic acid in the aliphatic chain. Most bile acids are conjugated with the amino acids glycine or taurine to give rise to bile salts (for example, glycocholic acid and taurocholic acid).



Cholic acid

Bile acids and salts are synthesized in the liver, stored in the gallbladder and secreted into the intestine through the bile duct after a fat-containing meal. Once in the intestine, most bile acids and salts are reabsorbed and return to the liver to be reused. This movement of molecules between the liver and the intestine is known as the **enterohepatic circulation**. The enterohepatic circulation conserves the bile acid pool, whereas the small fraction that escapes reabsorption and is excreted in feces contributes to the net elimination of cholesterol.

Steroid hormones

All steroid hormones derive from cholesterol. Still, only a small fraction of the total cholesterol is used for the synthesis of these hormones, because they are effective at low concentrations and do not need to be synthesized in large quantities.

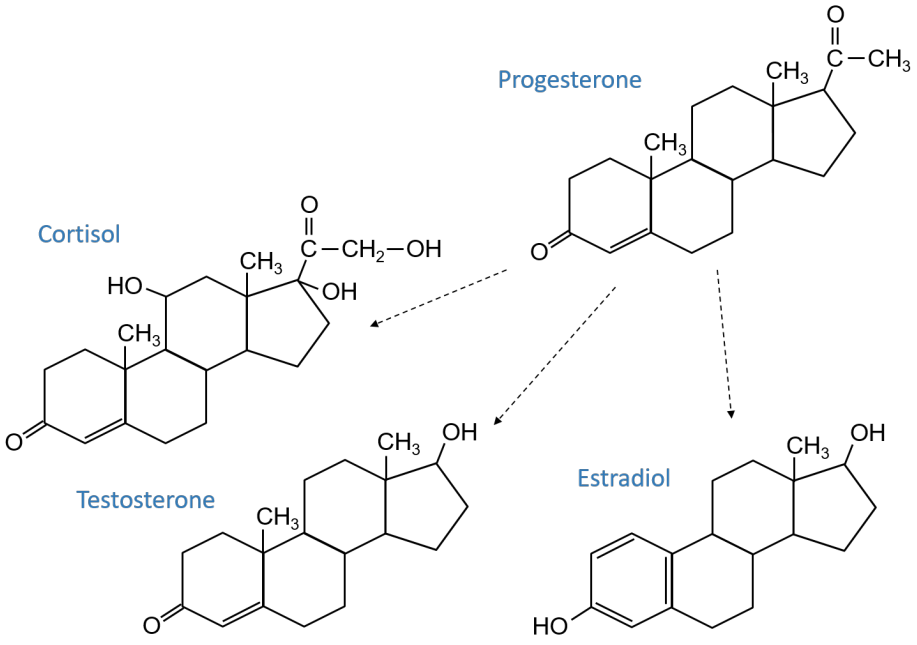
There are two main classes of steroid hormones based on where they are synthetized:

- **Corticosteroids**: they are synthesized in the adrenal cortex. There are two main types of corticosteroids:
    - **Mineralocorticoids**: regulate the ionic balance by promoting the reabsorption of ions in the kidneys.
    - **Glucocorticoids**: regulate carbohydrate metabolism, immune response and inflammation.

Examples are aldosterone (a mineralocorticoid) and cortisol (a glucocorticoid).

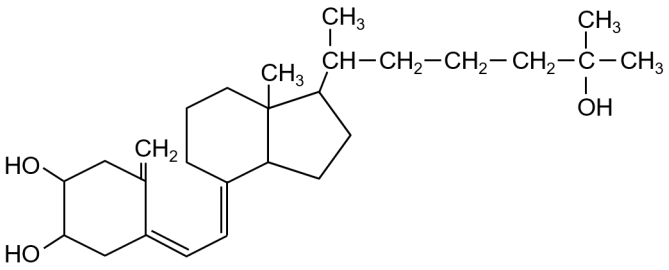
- **Sex hormones**: they are produced in male and female gonads and the placenta. They influence sexual development of males (**androgens**) and females (**estrogens**), or regulate the reproductive cycle and the maintenance of pregnancy (**progestogens**). Examples are testosterone (an androgen), estradiol (an estrogen) or progesterone (a progestogen).

All steroid hormones are synthesized from cholesterol through the common precursor pregnenolone, which is produced in adrenal glands, testes or ovaries. Pregnenolone is then converted into progesterone or into intermediates that give rise to corticosteroids (glucocorticoids and mineralocorticoids), and sex hormones (androgens and estrogens). Corticosteroids are recognized by a short and oxidized aliphatic chain, some hydroxyl groups and a carbonyl group at carbon 3. Androgens and estrogens lack the C17 side chain found in corticosteroids and progestogens. Estrogens also contain an aromatic ring.



Vitamin D_3

Vitamin D_3 (**cholecalciferol**) can be obtained from the diet, but it can also be synthesized from **7-dehydrocholesterol**, a cholesterol precursor. Therefore, strictly, it is not a vitamin, but a prehormone. It is not a steroid either, since, for its synthesis, the steroid core of 7-dehydrocholesterol is broken between C9 and C10. This ring opening occurs in the skin and is induced by **UV irradiation**. This is the reason why insufficient exposure to sunlight can cause vitamin D_3 deficiency. Still, cholecalciferol is not the active metabolite. Two mixed-function oxidases of the cytochrome P450 complex catalyze the hydroxylation of cholecalciferol. The first hydroxylation takes place in the liver at C25, and the second occurs in the kidney at C1, producing 1,25-dihydroxycholecalciferol (**calcitriol**).

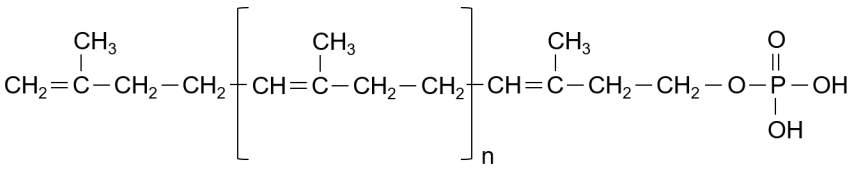


1,25-Dihydroxicholecalciferol

The active vitamin D_3 metabolite, calcitriol, promotes intestinal calcium and phosphate absorption and contributes to normal bone mineralization. **Vitamin D deficiency** leads to bone malformations. This is especially important in children, where the deficiency causes a disease known as **rickets**. In adults, the deficiency causes **osteomalacia**, a softening and weakening of the bones. Vitamin D_3 is often added as a dietary supplement in infants or when exposure to sunlight is limited.

Intermediates in cholesterol synthesis as precursors

Isopentenyl diphosphate, an intermediate in the synthesis of cholesterol, is a precursor of a large number of biomolecules that are generally known as **isoprenoids**. They include molecules such as dolichol, a lipid of the endoplasmic reticulum membrane whose function is to transport oligosaccharides to N-glycosylate proteins, or ubiquinone (coenzyme Q_{10}), an electron carrier in the mitochondria.



Dolichol phosphate

Prenylation, the covalent attachment of an isoprenoid to a protein, is a post-translational modification that allows proteins to anchor to the inner surface of cell membranes. An important example of prenylation occurs in the **Ras protein**, a protein involved in signal transduction and implicated in many human cancers. Ras is attached to a hydrophobic farnesyl group, which allows it to anchor to the plasma membrane. In cancer, the *RAS* gene is frequently mutated, generating a continuous transduction of signals and causing an uncontrolled division of cancer cells. Numerous efforts have been made to design drugs that prevent Ras prenylation, but so far with little success.

Other organisms produce isoprenoid molecules from intermediates in cholesterol synthesis, such as vitamin A (trans-retinol), vitamin E (α-tocopherol), vitamin K or terpenes (found in the perfumes and aromas of many plants).

Author: Ruben Pio

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